

SACHIN RAMESH JADHAV

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EDUCATION

Master of Engineering in AI & Robotics | University of Maryland, College Park, MD (Ivy league)(QS 218) CGPA : 3.71 | 2023 - 2025

Relevant coursework: Multimodal Foundational Models, Cloud Computing, Perception, Software Development, Reinforcement Learning, Path Planning, Programming in C++, Human Robot Interaction, Controls Systems

TECHNICAL SKILLS

Programming: Python | C++ | SQL | HTML | CSS | PostgreSQL | PySpark | PyTorch | TensorFlow | Spacy

Frameworks: Django | Flask | FastAPI | Open-CV | Astronomer Airflow | Agile | Langchain | CrewAi | Semantic Kernel | Autogen

Tools: AWS Sagemaker | Git | Terraform | Jenkins | MLflow | Docker | AWS Bedrock | Azure | JIRA | Kubernetes

Databases: MySQL | RDBMS | EMR | Rapid Miner | AWS Redshift | Snowflake | AWS Glue

PUBLICATIONS & CERTIFICATIONS

PLANRL: Combining Navigation and Imitation Learning Bootstrapping Reinforcement Learning, 2024 ISRR

[Link](#)

AWS Cloud Practitioner Certification

[Link](#)

PROFESSIONAL EXPERIENCE

Artificial Intelligence Research Intern | Nextstep Medical, USA

Aug 2024 - Present

Agentic Multimodal Chatbot for Medical Assistance

- Fine-tuned **BioViL-T** and **Mistral** multimodal LLMs on a private patient dataset using **LoRA**-based **PEFT** on **AWS SageMaker**.
- Developed a high-performance retrieval system with **FAISS** and **Colpali**, achieving sub-second query latency through optimized vector indexing and provided a comprehensive response through **prompt engineering**.
- Automated **HIPAA Compliant** infrastructure provisioning with **Terraform** remote states, reducing deployment cycles by 40%.
- Architected a serverless inference pipeline by deploying a **Flask** UI on **EC2**, triggering **AWS Lambda** via **S3** events, and routing responses through **API Gateway**, cutting costs by 30%.
- Enhanced radiologist support by improving patient data labeling and annotation with feedback-driven model refinement.

Predictive Modelling for IMU based Human Exoskeleton

- Spearheaded the development of an **attention-based RNN** architecture design with time series clustering for medical exoskeletons
- Fused data from Vicon motion capture and IMU sensors, with 40% improved terrain adaptation, reduced patient calibration time.

Transfer Q* AI Alignment

- Developed and implemented an AI alignment technique to optimize language model responses to human preferences using reward modeling (**RLHF**) and KV caching for enhanced response efficiency improving the performance by 19% over the baseline.

MLOps Engineer | Larsen & Toubro Technology Services, India (Sabic Account)

Jan 2022 - Jun 2023

Asset Health Monitoring

- Designed and implemented an **ETL pipeline** using **AWS Glue** (Crawler & ETL jobs) to process nested sensor data, optimize schema, and store partitioned Parquet files in **S3** for analytics
- Configured a **S3** Hook in an **Astronomer Airflow** infrastructure to deploy and monitor Machine Learning models in real-time.
- Optimized **anomaly detection models** to monitor turbine performance, utilizing **MLFlow** model registry on **DagsHub** to refine anomaly detection algorithms.

Data Science Engineer | Larsen & Toubro Technology Services, India (Valaris Account)

May 2021 - Jan 2022

Supplier Management

- Enhanced supplier evaluation by analyzing unstructured data from supplier reviews, contracts, and communication logs.
- Used **Named Entity Recognition (NER)** to extract key entities such as supplier names, contract terms, and performance metrics.
- Used Topic modeling techniques like **Latent Dirichlet Allocation (LDA)** to identify common themes in communication logs.
- Incorporated visualization tools such as bar charts, word clouds, and **Rapid Miner** to represent word impact on model's output.

Data Analyst | Larsen & Toubro Technology Services, India (Valaris Account)

Sept 2020 - Apr 2021

- Executed ETL processes to load data from various sources, such as **Excel, CSV, XML, and JSON files**, into the data warehouse.
- Integrated data from various sources, such as **RDS, DynamoDB, and API** endpoints, using **AWS Glue** and **Python** scripts.
- Implemented a time-series analysis pipeline using advanced statistical and machine learning techniques to forecast petroleum export trends, enhancing decision-making and strategy formulation.
- Maintained a data repository, creating and **managing Stored Procedures, Functions, Views, Triggers, and complex SQL queries** to support real-time analytics for predictive maintenance and asset health insights.
- Utilized this analysis to create dynamic **Tableau** dashboards, real-time monitoring and predictive insights into petroleum exports.

PROJECTS

Comparative Analysis of RAG-based LLMs for Medical Chatbot Applications

[Link](#)

- Evaluated the performance of **LLMs (Flan-T5-Large, LLaMA-2-7B, Mistral-7B)** in a medical QA context by finetuning and utilizing RAG.
- Achieved 57% test accuracy using the **Mistral-7B** model on the USMLE English dataset's MCQs. The model was fine-tuned on the MedMCQA dataset and leveraged a **RAG vector database**.

GAN Implementation | Pytorch, Reinforcement Learning, Actor-Critic

[Link](#)

- Implemented a hybrid learning system combining Behavioral Cloning (BC) and Q-learning in a **GAN** architecture, where the critic evaluates actions against expert behavior and environmental rewards, achieving robotic manipulation through adversarial training.

Brain Computer Interfaces | PySpark, Pandas, MLLib, EMR, Scikit-Learn

[Link](#)

- Optimized data infrastructure using statistical techniques and PySpark, applying feature engineering (PCA, low-pass filters) to evaluate **k-NN, SVM, Decision Forests, and Logistic Regression** models in MLLib and created an ensemble model in **AWS EMR**

Neural Network Development on Embedded Systems (Daikin Energy)

- Deployed Luxonis Oak-D-based real-time personnel detection and tracking system at **Daikin's Factory Unit**, achieving 40% faster **CNN** inference (0.691 mAP) through **ONNX optimization**, resulting in a 35% reduction in HVAC response time

DeepRRT-Star | Pytorch, Python, Numpy

[Link](#)

- Increasing computational efficiency by 60% using a **Deep Neural Network** channel by encoding the obstacle space using a contractive auto-encoder and training a 2D dataset to generate velocity and orientation of robot after each step for RRT* trajectory.